Chatbot POC
HRIPL

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Document Control

Revision history

Revision	Date	Record of Changes	Prepared By
1	30/01/2026	Version 0.1	Shreyash Kandarkar

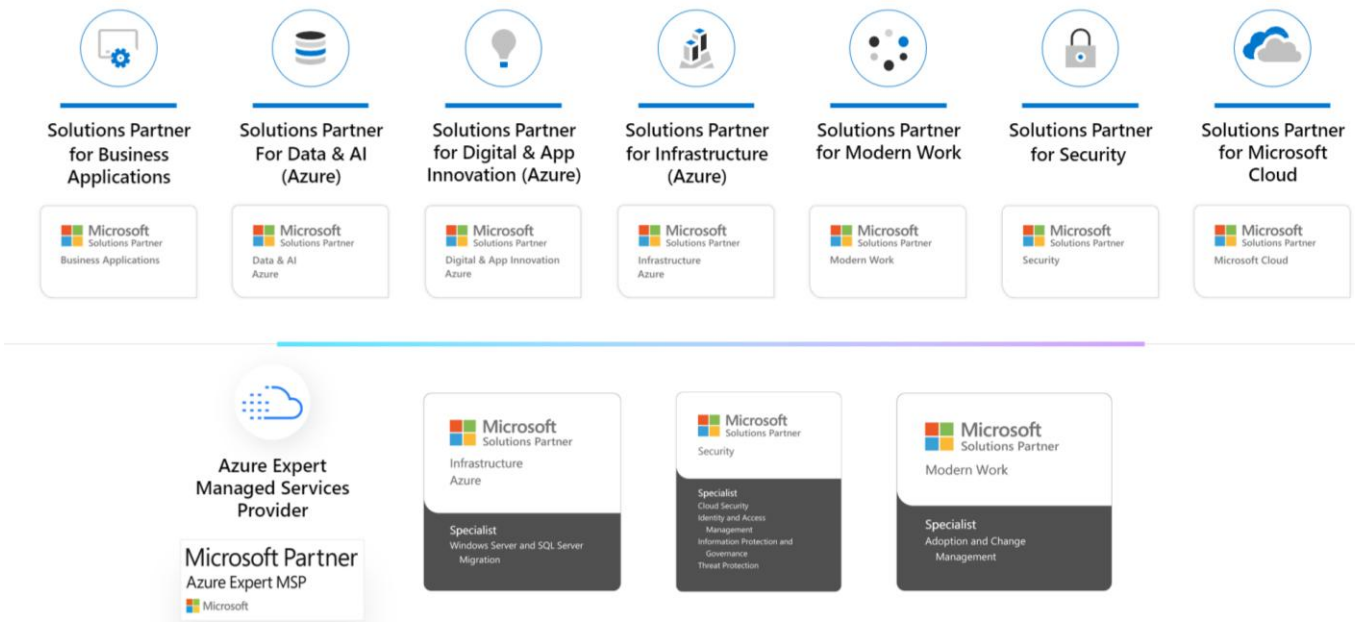
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1.1 Why atQor?

Aspect	atQor's Strengths	Value to Clients
Microsoft Expertise	Designated Microsoft Solutions Partner for Business Applications with deep expertise in Dynamics 365 and Power Platform.	Ensures seamless implementation and alignment with Microsoft's best practices and technology roadmap.
Proven Track Record	Recognized as the 2024 Microsoft India and South Asia Partner of the Year for Business Applications.	Demonstrates a strong history of delivering high-impact business solutions recognized by Microsoft.
Comprehensive Capabilities	Expertise across Dynamics 365 modules (Sales, Customer Service, Marketing), Power BI, and Power Automate.	Offers a unified platform approach to streamline operations, improve insights, and automate workflows.
Global Delivery Excellence	Balanced delivery model with certified teams across North America, India, and South Asia.	Ensures cost-effective solutions without compromising on quality or delivery timelines.
Industry Experience	Extensive experience in industries such as hospitality, financial services, manufacturing, and professional services.	Provides industry-specific insights to tailor solutions for unique business challenges.
Data and AI Leadership	Specialized in implementing advanced analytics and AI-driven insights within the Dynamics 365 ecosystem.	Empowers decision-making with actionable data and predictive intelligence.
Customization and Integration	Demonstrated expertise in migrating and integrating Dynamics 365 with tools like, Emailtopia, LinkedIn Sales Navigator, and PandaDoc through completed POCs and tailored demonstrations	Ensures seamless data flow between systems for enhanced efficiency and visibility.

Aspect	atQor's Strengths	Value to Clients
Customer-Centric Approach	Focused on delivering measurable business outcomes and fostering long-term partnerships.	Drives client success through tailored solutions and continuous support.
Scalable Solutions	Solutions built for scalability, leveraging Microsoft Azure and cloud-native technologies.	Future-proofs investments, ensuring flexibility for evolving business needs.
Recognition and Certifications	Teams certified across Dynamics 365, Power Platform, and Azure, with accolades for innovation and delivery quality.	Reflects credibility and ability to meet complex project requirements effectively.

Executive Summary

This project aims to deliver a production-ready AI ERP Agent that facilitates natural language queries against read-only SAP sales data, including master data, primary sales actuals, and sales planning targets stored in Microsoft Fabric. As part of this effort, a new data pipeline will be established to fetch data from the Silver Layer to Gold Layer in Microsoft Fabric, incorporating necessary calculations and aggregations currently absent in the source views.

The AI ERP Agent (deployed as a Microsoft Teams chatbot) will enable users to:

- **Query Product Master Data Instantly** — Access SKU attributes, material hierarchies (MG1-MG5), gross weight, UOM, and material group classifications through natural language questions without navigating SAP transactions.
- **Access Customer Intelligence On-Demand** — Retrieve customer profiles, GST/PAN details, territory assignments (Zone/State/SubZone, ASM/SO/SR), credit limits, and outstanding amounts conversationally.
- **Analyse Primary Sales Performance in Natural Language** — Get instant insights on actual sales by quantity, value, discounts, and taxes across dimensions like material, customer, plant, distribution channel, and time periods.
- **Compare Plan vs. Actual Achievement** — Measure sales performance against AOP and monthly plan targets by common dimensions (plant, distribution channel, month) to track achievement percentages and variances.

In addition to single-question capabilities, the AI ERP Agent will handle follow-up queries, maintaining conversation context for incremental refinements without requiring users to restate all details.

Supported Business (Process) Domains:

- **Material / Product Master Inquiry** – Material master attributes, material group, MG14–MG18 hierarchy, weight/UOM, size/UPC-like field, etc.
- **Customer Master Inquiry** – Customer master + sales area + GST/PAN + zone/state/subzone + credit/outstanding fields
- **Distribution Channel** – distribution channel codes and enable channel-based filtering.
- **Primary Sales (Actuals)** – Analyze actual sales (qty, value, discount, taxes) by invoice/material/customer/plant/channel/date.
- **Tax & Compliance from Sales (GST / HSN / E-Way)** – HSN, CGST/SGST/IGST % + amounts, EWayBill/EWayDate.
- **Pricing / Discount / MRP Performance (Sales-derived)** – Analyze SellPrice/InvoiceRate/MRP and discount fields from the sales fact.
- **Sales Planning** – Annual Operating Plan (AOP) Target Query – AOP targets (TotalSalesValuePrice, TotalQtyCV) by Year/Month, Zone/SubZone/State, DistributionChannel, PlantCode, EtprCode, SaleType.

- **Sales Planning** – Monthly Plan Target Query – Monthly targets (TotalSalesValuePrice, TotalQtyCV) by Year/Month, Zone/SubZone/State, DistributionChannel, PlantCode, EtprCode, SaleType.
- **Plan vs Actual (Achievement / Variance)** – Compare aggregated actual sales vs plan/AOP.

Proposed Scope

Supported Business (Process) Domains:

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- **Sales Planning – Monthly Plan Target Query –** Monthly targets (TotalSalesValuePrice, TotalQtyCV) by Year/Month, Zone/SubZone/State, DistributionChannel, PlantCode, EtprCode, SaleType.
- **Plan vs Actual (Achievement / Variance) –** Compare aggregated actual sales vs plan/AOP.

Data Pipeline & Management

- **Silver to Gold Layer Pipeline Implementation**
  - Implement a **data pipeline** within Microsoft Fabric to transform data from the **Silver Layer** (staging) to the **Gold Layer** (curated, business-ready for AI Agent).
  - **Calculations and aggregations** will be performed during this transformation, including:
    - Sales metrics: Actual sales (Qty, NetValue), discount analysis (TradeDisc%, TradeDisc), tax breakdowns (CGST/SGST/IGST amounts)
    - Plan vs Actual metrics: Achievement %, variance (Actual - Plan) by month/plant/channel
    - Pricing variance: MRP vs InvoiceRate variance, SellPrice analysis
    - Time intelligence: Monthly, YTD aggregations derived from Budat/Fkdat fields

        - HSN-wise and GSTIN-wise sales aggregations for compliance reporting
    - Data cleansing, null handling, and dimension mapping will be performed to ensure the data is usable by the AI ERP Agent.
    - Major schema redesigns or complex ETL workflows beyond the defined metrics remain out of scope.

- **Schema & Metadata**
    - Relevant schemas, including table/column definitions, primary/foreign key relationships, and business rules, will be documented.
    - Vector embeddings will be generated from these definitions and stored in Azure Cognitive Search to enhance T-SQL query generation accuracy within the AI ERP Agent.
    - This metadata is indexed to assist with semantic query parsing and domain identification (e.g., distinguishing Revenue queries from Capital Employed queries).

- **Access Restriction**
    - Only SELECT statements will be executed against the Fabric SQL Endpoint.
    - No data modifications (INSERT, UPDATE, DELETE) are permitted.
    - No new tables or views are created beyond what is needed for indexing metadata or storing temporary conversation state.

Multi-Agent Orchestration & Query Handling

- **Intent Classifier Agent**

  Purpose: The Intent Agent serves as the gateway classifier that analyzes user queries in natural language and determines the business domain, process type, and intent to route the request appropriately through the system.

  How It Works
    - Receives the raw user query in natural language and parses it to understand the core intent
    - Classifies the query into predefined business domains/categories (e.g., Production, Sales, Quality, Inventory)
    - Extracts multiple sub-questions if the user query contains compound requests
    - Returns a structured JSON response with classified questions and their associated process types for further processing

- **Functional Logic Agent**

  Purpose: The Functional Logic Agent acts as the business intelligence layer that maps user questions to the appropriate business rules, formulas, KPIs, and calculation logic required to answer the query.

  How It Works
  - Performs semantic search against a knowledge base of pre-defined business calculations and formulas
  - Identifies the relevant metrics, KPIs, and business logic associated with the user's question
  - Resolves dependencies between calculations (e.g., "Efficiency requires both Actual Output and Target Output")
  - Returns structured information including description, formula, required tables, lookup logic, and dependency mappings
  - Handles complex hierarchical calculations by identifying parent-child task relationships

- **Metadata Logic Agent**

  Purpose: The Metadata Logic Agent functions as the data dictionary resolver that translates business concepts and formulas into specific database schema elements including tables, columns, and data access patterns.

  How It Works
  - Searches the metadata repository to identify the exact tables and columns where required data resides
  - Validates that all required columns exist and are accessible for the requested query
  - Returns enriched task details with precise table names, column lists, and refined lookup logic

- **NLP-SQL/ NL Agent**

  Purpose: The NLP-SQL Agent serves as the query generation and execution engine that constructs valid SQL queries from the enriched metadata, executes them against the database, and translates raw results into meaningful natural language responses.

  How It Works
  - Receives enriched task information including tables, columns, formulas, and lookup logic from upstream agents
  - Generates syntactically correct T-SQL queries compatible with the target database

- Applies proper filtering, grouping, aggregation, and sorting based on the business logic requirements
- Executes the generated SQL queries against the database and retrieves result sets
- Handles parent-child task execution where child task results feed into parent calculations
- Interprets raw data and formulates human-readable answers explaining the results
- Maintains query history and execution data for audit and visualization purposes

- **Graph Agent**

  Purpose: The Graph Agent functions as the data visualization recommender that analyzes query results and determines the optimal chart type, data structure, and visual representation for presenting insights to end users.

  How It Works
  - Analyzes the structure and nature of the data returned from SQL query execution
  - Determines the most suitable visualization type based on data characteristics (bar chart, line graph, pie chart etc.)
  - Identifies meaningful comparisons, trends, or distributions that should be highlighted visually
  - Handles multiple result sets by generating appropriate visualization configurations for each

- **Follow-up / Context-Aware Agent**

  Purpose: The Follow-up Agent acts as the conversational memory layer that ensures users provide all necessary information before query processing, while also remembering what was discussed earlier to avoid repetitive questions.

  How It Works
  - Maintains awareness of the ongoing conversation and remembers previous questions, answers, and parameters shared by the user
  - Identifies when a user query is incomplete or missing required details
  - Automatically retrieves missing information from earlier conversation turns if the user already mentioned it before
  - Generates intelligent follow-up questions when essential details are not available in the current query or conversation history

- Decides whether to proceed directly with the query, ask for clarification, or use previously shared context
- Ensures smooth multi-turn conversations where users can refer back to earlier topics without repeating themselves (e.g., "What about last month?" continues from a prior plant-specific query)

Examples:

- User: *"Show me YTD revenue for SBG Alpha"*
- Bot: *[displays table]*
- User: *"Now show me just Division XYZ"* ← Bot understands to filter on Division XYZ within SBG Alpha context
- User: *"What about last year?"* ← Bot understands to show LY comparison

Monitoring

- **Query Logging**
  - All incoming user requests and outgoing SQL queries are logged for subsequent analytics on:
    - Usage patterns (most queried metrics, domains, time periods)
    - System performance (query execution time, error rates)
    - User adoption (active users, queries per user)
    - Will have separate interface as a web app form for Auditing & monitoring.

- **User Interface & Experience**
  - Microsoft Teams Chatbot
    - Users can query the AI ERP Agent through the Microsoft Teams interface.
    - The chatbot leverages Azure OpenAI for natural language understanding and generation.
    - Users can interact in a conversational manner without needing to know SQL.

- **Multiple Output Formats**
  - **Textual Summaries**: For quick numeric or textual answers.
    - "Your FTM Revenue is 1,781 Cr, which is 98% of AOP (1,815 Cr). You are -34 Cr below target."
  - **Tabular Displays**: For itemized or comparative data, such as:

        - Plan vs Actual Table — Month, plant, channel, actual sales, plan target, achievement %, variance
        - Top/Bottom Performers Table — Top 10 SKUs by net value, bottom 10 by achievement % ,Capital Employed breakdown (Gross CE, Net CE, LY vs CY)
        - Primary Sales Performance Table — Plant, material, customer, quantity, net value, discount, tax amounts
        - Tax & Compliance Table — Invoice number, HSN code, CGST/SGST/IGST amounts, E-way bill number/date
        - Note: Microsoft Teams response is limited to 40 KB of characters, so large tables may require summarization or pagination.
    - **Graphical Visualizations**: For intuitive trend analysis, performance comparison, and distribution insights.
- **Supported chart types include:**
    - **Bar Charts** – category-wise comparisons
    - **Line Charts** – time-series trends
    - **Pie Charts** – proportional distribution
- **Visualization Behaviour:**
    - Graphs shall be generated dynamically based on query context.
    - Visual outputs may be delivered as inline images or downloadable artifacts within Microsoft Teams or integrated UI channels.
    - For large datasets, charts shall automatically aggregate data to maintain readability and performance.

Assumptions

- **Read-Only Operations**: No data modifications (INSERT, UPDATE, DELETE) are required.
- **Limited Multi-Turn Memory**: The system offers **short-term chat context** (3-5 turns); extensive long-term conversation memory is not part of this phase.
- **English-Only**: All user queries and responses are in **English**.
- **Data Availability**:
    - Silver Layer data is **complete and accurate** for the defined scope.
    - Data is **regularly updated** (e.g., daily, weekly) to reflect current business performance

- **Standard Chatbot UI**: The production UI leverages native **Microsoft Teams** functionality, with only basic styling for tables and messages.
- **Schema Scope**: Only the 6 views identified in the provided schema documentation will be considered for this engagement:
    - Material master (material codes, descriptions, specifications)
    - Customer Master (contact details, addresses, GST information, credit limits)
    - Primary Sales (invoice details, quantities, pricing, discounts)
    - Channel Master (Retail, Modern Trade, Professional, Institutional, and E-commerce channels)
    - Annual Plan (Annual sales targets by month, zone, state, channel)
    - Monthly Plan (Monthly sales targets broken down by geography (zone/state), distribution channel)
- **All Data is Read-Only**: No inserts, updates, or deletes.
- **Appropriate Azure Subscription/Resources** are available:
    - Microsoft Fabric with sufficient capacity
    - Azure OpenAI, Azure AI Search, Azure App Service, Azure Key Vault
- **Microsoft Teams** is the chosen interface for end-user queries.
- **MS Teams Response Limit**: Response is limited to **40 KB of characters**. Large tables may require summarization or pagination.
- **Text-Based Input Only**: Only text-based queries will be considered for this engagement (no voice, images, or file uploads).
- **AI Response Formats**: Text, table & Graph (Line, Bar, Pie) format will be generated for this engagement (no complex visualizations, Power BI embeds). & Graph will not be responsive & embedded as Image within Teams Platform.
- **Success Criteria**: Will be defined during the initial kick-off call with the customer (e.g., 90% query success rate, <30 second response time, 95% user satisfaction).
- **Business Definitions**:
    - **Material Hierarchy** — MG1–MG5 levels map to Brand/Division/Product Group/Sub-category (to be confirmed during requirements phase)
    - **Plan Matching** — Plan vs Actual limited to Plant, Distribution Channel, and Month dimensions only; EtprCode has no mapping to Material/MG fields;

Zone/State/SubZone alignment between customer master and plan views must be validated

- **BillTo & Channel Coverage** — Null BillTo records (stock transfers) exclude customer/territory analysis; unmapped distribution channels (e.g., channel 90) have no description available
- **Data Scope** — Limited to Master Data + Primary Sales + Plan targets only; excludes FI/AR/AP aging, Inventory/stock movements, and Order-to-Cash pipeline

- **Data Quality**:
  - Silver Layer data has been validated for completeness and accuracy by the business
  - Any data quality issues (missing values, incorrect mappings) will be flagged for remediation before pipeline implementation
- **No Advanced Analytics**:
  - AI/ML-based forecasting, predictive analytics, and optimization recommendations are **out of scope** for this phase
  - The agent provides **descriptive and diagnostic analytics** (what happened, why it happened) only
- MASTER DATA LIMITATIONS:
  - **Material Hierarchy** — MG1–MG5 mapping to business categories (Brand/Division/Product Group) to be confirmed during requirements phase
  - **Customer Territory** — ASM/SO/SR fields exist but zone/state/subzone alignment with plan geography requires validation
  - **Distribution Channel** — Unmapped channels in sales actuals (e.g., channel 90) will have no description available from vw_DistrChannel.

- SALES ANALYSIS LIMITATIONS:
  - **BillTo Null Scenarios** — Stock transfer transactions (ZSTO) with blank BillTo cannot be analyzed by customer, territory, or GSTIN dimensions
  - **Customer Outstanding** — Limited to OutAmt_OutstandingAmount and CreditLimit_CreditLimit fields in customer master; detailed AR aging, payment history, and invoice-level collections are not available
  - **E-way Bill Tracking** — E-way bill number and date available but compliance threshold rules require business confirmation

- PLAN vs ACTUAL CONSTRAINTS:
  - **Safe Matching Dimensions** — Plan vs Actual limited to Month/Year, Plant (Werks ↔ PlantCode), Distribution Channel (Vtweg ↔ DistributionChannel), and SaleType only

    - **EtprCode Limitation** — EtprCode exists in plan views but has no mapping to Material/MG fields; material-level plan matching not possible
    - **Territory Matching** — Zone/State/SubZone plan dimensions may not align with customer master territory fields; reliability requires validation
    - **Price/Mix/Volume Variance** — Requires data not in current model; only overall Achievement % and Variance (Actual - Plan) available
- TIME PERIOD CONSTRAINTS:
    - Time intelligence supports: FTM, QTD, YTD, FY
    - Historical trend analysis limited to data available in Silver Layer
    - Real-time data updates depend on Silver Layer refresh schedule
- RESPONSE FORMAT CONSTRAINTS:
    - Microsoft Teams bot responses limited to 40 KB text/tables
    - Text Input is only considered in English language only.
    - Large result sets require pagination or summarization

These assumptions will be revisited at the project kick-off and any deviations will be addressed (for example, if data is not readily available or if enabling an Azure service is delayed, the plan may be adjusted).

Out of Scope

The following are **explicitly excluded** from this engagement:

- **Data & Analytics:**
    - **Price/Mix/Volume Variance Analysis** – Not in data model; requires new data sources
    - **Material Consumption Tracking** – Requires integration with inventory/ERP system
    - **Resource Utilization (FTE/Hours)** – Requires integration with HRMS/timesheet system
    - **ROI by Project** – Requires investment/CapEx data not in current model
    - **Billing Status by Project** – Not in data model
    - **Role Base access control** on Data Level (Row Level Security) will be considered as out of scope.
- **Technical:**
    - **ETL/ELT for More Than 6 Views** – Additional tables require separate scoping
    - **Complex Long-Term Memory** – Storing extended historical context over multiple chat sessions (only short-term 3-5 turn memory is supported)
    - **Write/Update Operations** – No database inserts, updates, or deletions handled by the agent

- **Advanced Custom UI** – Beyond native Microsoft Teams capabilities (e.g., custom web portal, mobile app)
- **Multi-Lingual Support** – Responses are **English-only**
- **Desktop/Mobile App Development** – No separate application interfaces are included
- **Processing Unstructured Data** – PDF, Excel files, images, voice input are **not supported**
- **Audit & Compliance Dashboard (Separate Web App)** – No dedicated web interface for query logs, authentication events, or compliance reporting
- **Performance Metrics Dashboard** – No separate interface for tracking response time, query success rate, or user satisfaction KPIs

- **Advanced AI/ML:**
  - **AI/ML Forecasting** – Predictive models for Q4 revenue forecasting, trend extrapolation (requires ML model development)
  - **Risk Alerts with Scoring** – Beyond simple rule-based alerts (e.g., "Delayed" flag); advanced risk scoring requires ML
  - **Recommendation Engine** – Optimization algorithms for resource reallocation, project prioritization (requires ML/optimization models)
- **Integration:**
  - **Integration with Non-Fabric Systems** – Direct integration with SAP, Oracle, or other ERP systems (data must be available in Fabric Silver Layer)
  - **Real-Time Streaming Data** – Agent queries near-real-time data in Fabric; sub-second streaming is not supported

Pre-requisites

- **Operational Microsoft Fabric Workspace:**
  - Access to **Silver Layer** data.
  - Permissions to create Additional **Gold Layer** tables/views and Fabric pipelines
  - **Fabric SQL Endpoint** provisioned with read-only access for the AI Agent
- **Azure Resources:**
  - **Azure OpenAI** service with GPT-4.1 or GPT-5.2 model deployment
  - **Azure AI Search** (or equivalent) for metadata indexing
  - **Azure Key Vault** for secure credential storage

  - **Azure App Service** for hosting the orchestrator

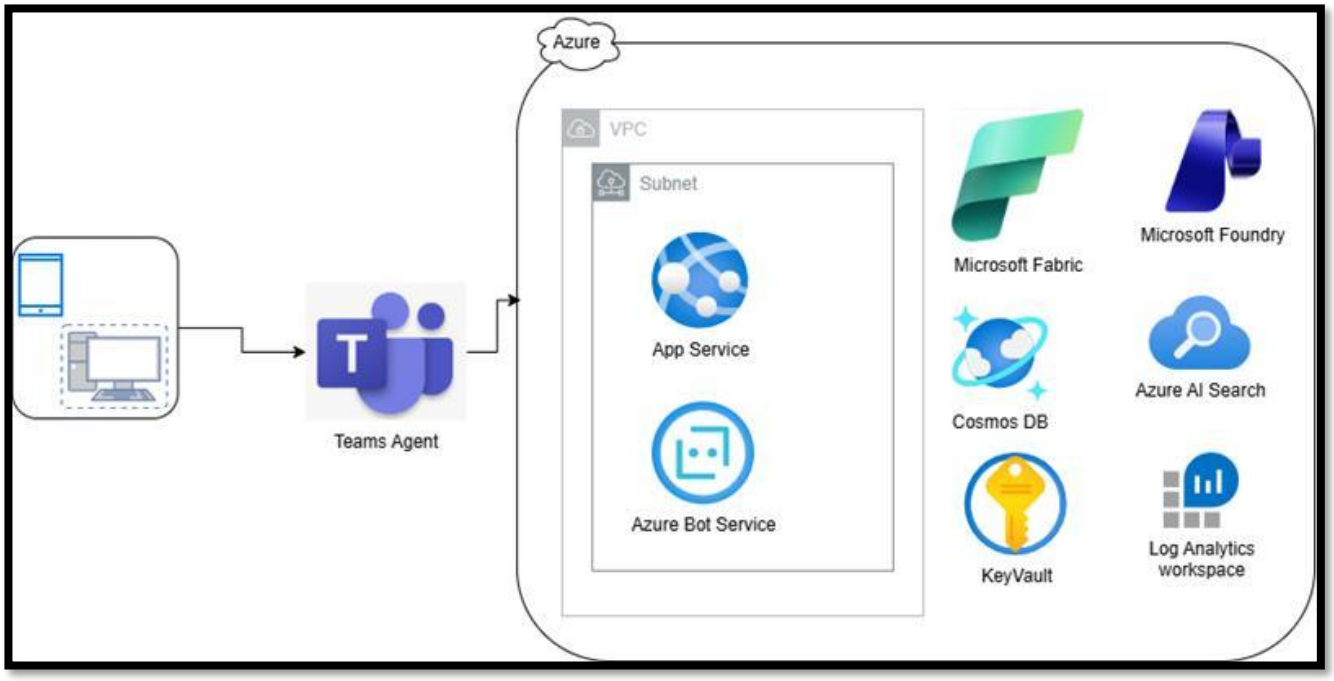
- **Microsoft Teams Bot Registration:**
  - Verified permissions and capacity within **Microsoft Teams** to deploy and test the bot
  - App registration in **Microsoft Entra ID** with appropriate API permissions
- **Adequate Azure OpenAI Quota:**
  - ≥ **450K tokens/min** recommended for production usage (to be sized based on expected concurrency)
- **Data Understanding:**
  - **Confirm Material Hierarchy Mapping** — Validate MG14–MG18 hierarchy alignment with business categories (Brand/Division/Product Group/Sub-category)
  - **Validate Plant and Territory Mapping** — Confirm Plant Code (Werks/PlantCode) formatting consistency between actuals and plan views; verify Zone/State/SubZone alignment between customer master and plan dimensions
  - **Clarify Distribution Channel Usage** — Obtain and implement the customer's internal channel mapping specification that defines business channels.
  - **Define Plan vs Actual Matching Rules** — Establish safe join dimensions (Month/Plant/Channel) and document limitations for EtprCode and territory-based matching
- **Test Data & Sample Queries:**
  - Business users provide **sample queries** representing real-world use cases which will be facilitate to set the Success Criteria.

Deliverables

- **Production AI ERP Agent**:
  - Deployed as **Microsoft Teams chatbot** with natural language query capabilities
  - Provides **short-term follow-up interactions** (3-5 turn context)
  - Enforces **read-only access** with role-based security (via Entra ID)
- **Gold Layer Data Pipeline:**
  - **Fabric pipeline** (Notebook or Data Flow) to transform Silver → Gold with:
    - **Master Data Views** — Material master (MG1–MG5 hierarchy, weight/UOM), Customer master (GST/PAN, territory, credit/outstanding), Distribution channel mapping

        - **Sales Actuals Metrics** — Quantity, NetValue, tax calculations (CGST/SGST/IGST), discount metrics (TradeDisc%), pricing variance (MRP vs InvoiceRate)
        - **Plan/AOP Targets** — TotalSalesValuePrice, TotalQtyCV by Zone/State, Plant, Channel with dimension alignment
        - **Plan vs Actual Calculations** — Achievement % = (Actual / Plan) × 100, Variance = Actual - Plan, Monthly and YTD aggregations by Plant/Channel/Month
    - Scheduled refresh (daily/weekly as per business requirement)
- **Metadata & Vector Index:**
    - **Azure Cognitive Search** index populated with:
        - Table/column definitions for Gold Layer schema
        - Business glossary
        - Vector embeddings for semantic search and query disambiguation
    - Documentation of schema relationships and calculation logic
- **Secure Teams Chatbot:**
    - Integrated with **Microsoft Entra ID** for SSO and MFA
    - Integrated with **Azure OpenAI** for natural language understanding and generation
    - Error handling and clarification prompts for ambiguous queries
- **Contextual Chat Feature:**
    - Limited **multi-turn dialogue** with session memory (3-5 recent turns)
    - Context-aware follow-up handling (e.g., "Now show me last year", "What about Division XYZ?")

Architecture Diagram



Success and Acceptance Criteria

Success Criteria

The following success criteria will be **finalized during the initial kick-off call** with HRIPL team:

- **Query Success Rate**: ≥ **90%** of user queries are successfully translated to SQL and return correct results (measured during UAT and first month post-go-live)
- **Response Time**: ≥ **90%** of queries return results within **30 seconds** (excluding complex aggregations or large result sets)
- **Context Accuracy**: ≥ **85%** of follow-up queries correctly interpret conversational context without requiring the user to restate information
- **Calculation Accuracy**: **100%** of calculations in Gold Layer match business requirements (validated during UAT by Finance team)
- **Availability**: ≥ **99%** uptime during business hours (excluding planned maintenance)

Acceptance Criteria

The project will be considered **complete** and ready for **production go-live** when:

- All **9 business domains** are implemented and tested:
    - Material / Product Master Inquiry
    - Customer Master Inquiry
    - Distribution Channel
    - Primary Sales (Actuals)
    - Tax & Compliance from Sales
    - Pricing / Discount / MRP Performance (Sales-derived)
    - Sales Planning – (AOP) Target Query
    - Sales Planning – Monthly Plan Target Query
    - Plan vs Actual (Achievement / Variance)
- **Gold Layer pipeline** is deployed and running on schedule (daily/weekly refresh)
- **UAT is successfully completed** with sign-off from Finance and Operations teams:
    - All test cases passed (≥90% success rate)
    - No critical or high-priority bugs outstanding
- **Contextual chat** and follow-up queries are working as expected (validated in UAT)

Timeline

Phase	Feature / Module	Description
Inception & Requirements	Requirement Gathering, Planning & Infrastructure Setup	Requirement clarification (Margin definition, Business = Division, Customer analysis priority), Infrastructure setup (Fabric workspace, Azure OpenAI, etc.)
Solution Design	Metadata & Schema Setup	Create aggregated views, gold layer schema, table relationships, create business glossary for agents.
Development	Refining Gold Layer	Implement and refine the Gold layer with calculated metrics (sales actuals, tax breakdowns, pricing variance, and Plan/AOP targets)
Development	AI Architecture	Define database schema and relationships, document KPIs in a business glossary, and generate vector embeddings for this metadata to store in Azure AI Search for semantic retrieval.
Development	Metadata & Functional Vector Store Setup	
Development	Orchestrator & Intent Agent	Build the main orchestrator on Azure App Service, enable intent routing across 9 business domains, and implement chain-of-thought flow between agents.
Development	NL-to-SQL & Logic Agents	**Functional Agent:** Logic to retrieve formulas/KPIs **Metadata Agent:** Logic to resolve table/column names **NLP-SQL Agent:** Prompt engineering for accurate T-SQL generation

Development	Graph & Visualization Agent	Logic to auto-select Chart Type (Bar/Line/Pie) based on data shape
Development	Context & Follow-up Agent	Implement conversational memory (3–5 turns) and merge previous context with the current query.
Development	Teams Bot Integration	Configure Teams bot with Azure OpenAI, implement bot framework, secure API calls with managed identity, Downloadable Excel
Functional Testing	UAT	End-to-end testing with business users across all 9 domains, test conversational context, validate calculations, performance testing
Functional Testing	UAT Bug Fixing	Address issues identified during UAT, refine query logic, improve response formatting
Go-Live	Deployment & Documentation	Production rollout, user guides (how to query the bot), architecture documentation

The estimated time for the completion of POC is 7-8 weeks

Estimated Azure Consumption

Service category	Service type	Region	Description	Estimated monthly cost	Remarks
AI + Machine Learning	Azure OpenAI	South India	Language Models, GPT 4.1, Standard (On-Demand), Global, 5,500 x 1,000 input tokens, 0 x 1,000 cached input, 137.5 x 1,000 output tokens	₹1,088.77	Below is the consideration of Intent Identification Agent 50 queries/day × 22 days = 1,100 queries/month Considering 5 users Input Tokens= 1,000 × 1,100 × 5 = 5,500,000 input tokens / month Output Tokens = 25 × 1,100 × 5 = 137,500 output tokens / month
AI + Machine Learning	Azure OpenAI	Southeast Asia	Embedding Models, Text-Embedding-3-Large, 16,501 x 1,000 Tokens	₹255.38	Below is the consideration of Followup Agent 50 queries/day × 22 days = 1,100 queries/month Considering 5 users Considered input Total = 3000 * 1,100 * 5 = 16,500,000

AI + Machine Learning	Azure OpenAI	South India	Language Models, GPT 4.1, Standard (On-Demand), Global, 11,000 x 1,000 input tokens, 0 x 1,000 cached input, 11,000 x 1,000 output tokens	₹9,897.94	Below is the consideration of Functional Logic Agent 50 queries/day × 22 days = 1,100 queries/month Considering 5 users Input Tokens= 2000 × 1,100 × 5 = 11,000,000 input tokens / month Output Tokens = 2000 × 1,100 × 5 = 11,000,000 output tokens / month
AI + Machine Learning	Azure OpenAI	South India	Language Models, GPT 4.1, Standard (On-Demand), Global, 11,000 x 1,000 input tokens, 0 x 1,000 cached input, 2,750 x 1,000 output tokens	₹3,959.18	Below is the consideration of Metadata Logic Agent 50 queries/day × 22 days = 1,100 queries/month Considering 5 users Input Tokens= 2000 × 1,100 × 5 = 11,000,000 input tokens / month Output Tokens = 500 × 1,100 × 5 = 2,750,000 output tokens / month

AI + Machine Learning	Azure OpenAI	South India	Language Models, GPT 4.1, Standard (On-Demand), Global, 11,000 x 1,000 input tokens, 0 x 1,000 cached input, 4,400 x 1,000 output tokens	₹5,146.93	Below is the consideration of NLP-SQL/ NL Agent 50 queries/day × 22 days = 1,100 queries/month Considering 5 users Input Tokens= 2000 × 1,100 × 5 = 11,000,000 input tokens / month Output Tokens = 800 × 1,100 × 5 = 4,400,000 output tokens / month
AI + Machine Learning	Azure OpenAI	East US	Language Models, GPT 4.1, Standard (On-Demand), Global, 2,744 x 1,000 input tokens, 0 x 1,000 cached input, 550 x 1,000 output tokens	₹889.73	Below is the consideration of Graph Agent 50 queries/day × 22 days = 1,100 queries/month Considering 5 users Input Tokens= 500 × 1,100 × 5 = 2,750,000 input tokens / month Output Tokens = 100 × 1,100 × 5 = 550,000 output tokens / month
Web	Azure AI Search	South India	Basic, 2 Unit(s), 24 Hours	₹574.44	

Databases	Azure Cosmos DB	South India	Azure Cosmos DB for NoSQL (formerly Core), Standard provisioned throughput (manual), Always-free quantity disabled, Pay as you go, General Purpose, Single Write- South India (Write Region), 1,200 RUs x 730 Hours, 0 GB transactional storage, Analytical storage disabled, 2 copies of periodic backup storage, Dedicated Gateway not enabled	₹6,542.36	
Compute	App Service	Central India	Premium V3 Tier; 1 P1V3 (2 vCPU(s), 8 GB RAM, 250 GB Storage) x 730 Hours; Linux OS; 0 SNI SSL Connections; 0 IP SSL Connections; 0 Custom Domains; 0 Standard SLL Certificates; 0 Wildcard SSL Certificates	₹10,772.56	
Analytics	Azure AI Bot Service	South India	S1 tier, 0 messages in Premium Channels	₹0.00	For Standard channel, in S1 sku Unlimited messages are included.
			Total	**₹39,127.28**	

Sign-Off

The Scope would be binding and final for the complete development under this contract. Any development/implementation/consultancy which is not covered under the Project Scope would be considered as a Change Request and extra charges would be levied as per actual.

Signed for and on behalf of:	Signed for and on behalf of:
atQor	HRIPL
Signed:	Signed: .....
Name:	Name: .....
Title:	Title: .....
Date:	Date: .....